

MULTIPLE LINEAR REGRESSION MODEL

Q1) Prove Theorem 2-(b), (c)

$$X \in \mathbb{R}^{n \times (p+1)}, X \text{ is full rank}$$

(a) $\text{rank}(X) = p+1$

(b) Number of observations must be greater than that of predictors

(c) The $(p+1) \times (p+1)$ real matrix $X^T X$ is invertible

(b) $n \geq p+1$

$$\text{row-rank}(A) = \text{column-rank}(A)$$

if $n < p+1$, $\text{row-rank}(A) > \# \text{ of rows in } A \Rightarrow \text{contradiction}$

$$\therefore \text{row-rank}(A) \leq \# \text{ of rows in } A$$

(c) $\text{rank}(X) = \text{rank}(X^T X)$

$$\therefore \text{rank}(X^T X) = p+1 \Rightarrow X^T X \text{ is a square matrix and is full rank}$$

$$\Leftrightarrow X^T X \text{ is non-singular (= invertible)}$$

Q2) Prove remark: the design matrix X must be full column rank in order for the model to be identifiable

$$y = X\beta + \varepsilon, X = [1 \ x_1 \ \dots \ x_p] \in \mathbb{R}^{n \times (p+1)}, \beta = (\beta_0 \ \dots \ \beta_p)^T \in \mathbb{R}^{(p+1) \times 1}$$

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T \in \mathbb{R}^n$$

$$(\text{model 1}) : y = X\beta + \varepsilon, \varepsilon_i \stackrel{iid}{\sim} (0, \sigma^2)$$

$$(\text{model 2}) : y' = X\beta' + \varepsilon, \varepsilon_i \stackrel{iid}{\sim} (0, \sigma'^2)$$

Suppose that y and y' are the same model.

Then, the distribution of y and y' are identical, meaning, $\sigma^2 = \sigma'^2$

$$\therefore y = y' \Leftrightarrow X\beta = X\beta' \Rightarrow X^T X\beta = X^T X\beta' \Rightarrow [X^T X]^{-1} X^T X\beta = [X^T X]^{-1} X^T X\beta'$$

$$\Rightarrow \beta = \beta'$$

$\therefore$ The design matrix must be full rank (= invertible) for the model to be identifiable.

Method of Estimation

Q3) Prove Theorem 3

$$\hat{\beta} = \underset{\beta \in \mathbb{R}^{p+1}}{\text{argmin}} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2 = \underset{\beta \in \mathbb{R}^{p+1}}{\text{argmin}} \|y - X\beta\|^2$$

Since we know that $C_X = \{X\beta : \beta \in \mathbb{R}^{p+1}\}$

it follows that $X\hat{\beta} = \hat{z}$ where $\hat{z} = \underset{z \in C_X}{\text{argmin}} \|y - z\|^2$

Now, show that $X\hat{\beta} = \hat{z}$ where $\hat{z} = \underset{z \in C_X}{\text{argmin}} \|y - z\|^2$

* note 1: there are two approaches to solve this question

* note 2: Show that the Hessian matrix is positive definite.

(Approach #1)

$$\hat{z} = \underset{z \in C_X}{\text{argmin}} \|y - z\|^2,$$

$\hat{z}$ is the orthogonal projection of y onto the Column Space of X

$\therefore \hat{z} = \Pi_X y = X(X^T X)^{-1} X^T y$ due to the "closest vector property".

$$X\hat{\beta} = \hat{z} \therefore X\hat{\beta} = X(X^T X)^{-1} X^T y \Rightarrow X^T X\hat{\beta} = X^T X(X^T X)^{-1} X^T y$$

$$\Rightarrow (X^T X)^{-1} X^T X\hat{\beta} = (X^T X)^{-1} X^T X(X^T X)^{-1} X^T y$$

$$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$$

(Approach #2)

$$L(\beta) = \|y - X\beta\|^2 = (y - X\beta)^T (y - X\beta) = y^T y - 2y^T X\beta + \beta^T X^T X\beta$$

$$\partial L(\beta) / \partial \beta = -2y^T X + 2X^T X\beta$$

$$[\partial L(\beta) / \partial \beta]_{\beta = \hat{\beta}} = -2y^T X + 2X^T X\hat{\beta} = 0$$

$$\Leftrightarrow X^T X\hat{\beta} = X^T y$$

$$\Leftrightarrow \hat{\beta} = (X^T X)^{-1} X^T y$$

$$\frac{\partial^2 L}{\partial \beta^2} = 2X^T X \quad \forall c \in \mathbb{R}^{(p+1)}$$

$$c^T X^T X c = \|Xc\|^2 \geq 0 \therefore X^T X \text{ is non-negative definite}$$

+ $X^T X$ is invertible: n.n.d & invertible $\Rightarrow X^T X$ is positive definite

$\Rightarrow \hat{\beta}$ is the minimum critical point

Q4) Prove Theorem 4

$$(a) E[\hat{\beta}] = \beta$$

$$(b) \text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

$$(a) \hat{\beta} = (X^T X)^{-1} X^T y, \therefore E[(X^T X)^{-1} X^T y] = (X^T X)^{-1} X^T E[y] \\ = (X^T X)^{-1} X^T X \beta = \beta$$

$$(b) \text{Var}((X^T X)^{-1} X^T y) = (X^T X)^{-1} X^T \text{Var}(y) X (X^T X)^{-1} \\ = (X^T X)^{-1} X^T (\sigma^2 I_p) X (X^T X)^{-1} \\ = \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \\ = \sigma^2 (X^T X)^{-1}$$

Q5) Prove Theorem 5

Decompose $X = (\mathbb{1} \ X_1)$ and define $X_{1,\perp} = X_{\mathbb{1}} - \Pi_{\mathbb{1}} X_1$. Prove the following:

$$(a) C(X_{1,\perp}) \text{ is orthogonal to } C(\mathbb{1})$$

$$(b) C(X) = C(\mathbb{1}) + C(X_{1,\perp})$$

$$(a) X_{1,\perp} = X_1 - \Pi_{\mathbb{1}} X_1 \\ \langle X_1 - \Pi_{\mathbb{1}} X_1, \mathbb{1} \rangle = X_1^T \mathbb{1} - (\Pi_{\mathbb{1}} X_1)^T \mathbb{1} \\ = X_1^T \mathbb{1} - X_1^T \Pi_{\mathbb{1}} \cdot \mathbb{1} \\ = X_1^T \mathbb{1} - X_1^T \mathbb{1} (\mathbb{1}^T \mathbb{1})^{-1} \mathbb{1}^T \mathbb{1} \\ = X_1^T \mathbb{1} - X_1^T \mathbb{1} \\ = 0$$

$$\therefore \mathbb{1} \text{ and } X_{1,\perp} \text{ are orthogonal} \Leftrightarrow C_1 = \mathbb{R}, C_2 \in \mathbb{R}^p, \mathbb{1} C_1 \text{ and } X_{1,\perp} C_2 \text{ are} \\ \text{orthogonal} \\ \Leftrightarrow \mathbb{1} C_1 = C(\mathbb{1}) \text{ and } X_{1,\perp} C_2 = C(X_{1,\perp}) \\ \therefore C(\mathbb{1}) \text{ and } C(X_{1,\perp}) \text{ are orthogonal}$$

$$(b) C(X) = C(\mathbb{1}) + C(X_{1,\perp})$$

$$\text{WTS: } \textcircled{1} C(X) \subset C(\mathbb{1}) + C(X_{1,\perp})$$

$$\textcircled{2} C(\mathbb{1}) + C(X_{1,\perp}) \subset C(X)$$

$$\textcircled{2} \mathbb{1} \in C(X) \text{ is trivial}$$

$$X_{1,\perp} = (x_1 \dots x_p) - \Pi_{\mathbb{1}} (x_1 \dots x_p) \\ = (x_1 - \Pi_{\mathbb{1}} x_1) \dots (x_p - \Pi_{\mathbb{1}} x_p) \\ = (x_1 - \bar{x}_1 \mathbb{1}) \dots (x_p - \bar{x}_p \mathbb{1})$$

$$\Pi_{\mathbb{1}} x = (x_1 + \dots + x_n) = (\bar{x} + \dots + \bar{x}) = \bar{x} \mathbb{1}$$

$$\therefore C(X_{1,\perp}) : X \text{ 에서 } C(\mathbb{1}) \text{ 을 빼준 것이기 때문에 } C(X_{1,\perp}) \in C(X)$$

$$\therefore C(\mathbb{1}) + C(X_{1,\perp}) \in C(X)$$

$$\textcircled{1} C(X) \subset C(\mathbb{1}) + C(X_{1,\perp})$$

We will show that the $\dim(\text{LHS}) = \dim(\text{RHS})$

$$\dim(C(\mathbb{1}) + C(X_{1,\perp})) = \dim(C(\mathbb{1})) + \dim(C(X_{1,\perp})) - \dim(C(\mathbb{1}) \cap C(X_{1,\perp}))$$

X is full rank

$$\therefore x_1 - \bar{x}_1 \mathbb{1}, x_2 - \bar{x}_2 \mathbb{1}, \dots, x_p - \bar{x}_p \mathbb{1} \text{ are linearly independent.}$$

$$(\text{Reasoning}): a_1(x_1 - \bar{x}_1 \mathbb{1}) + \dots + a_p(x_p - \bar{x}_p \mathbb{1}) = 0 \\ \Rightarrow a_1 x_1 + \dots + a_p x_p - (a_1 \bar{x}_1 + \dots + a_p \bar{x}_p) \mathbb{1} = 0$$

$$\text{Since the column vectors of } X \text{ are linearly independent,} \\ a_1 x_1 + \dots + a_p x_p = 0 \Leftrightarrow a_1 = \dots = a_p = 0$$

$$\therefore a_1 x_1 + \dots + a_p x_p = 0 \text{ and } a_1 \bar{x}_1 + \dots + a_p \bar{x}_p = 0$$

$$\therefore \dim(C(X_{1,\perp})) = p$$

$$\dim(C(\mathbb{1}) \cap C(X_{1,\perp})) = \dim(\{0\}) = 0$$

$$\therefore \dim(C(\mathbb{1}) + C(X_{1,\perp})) = 1 + p = \dim(C(X)) = \text{rank}(X)$$

$$\therefore C(\mathbb{1}) + C(X_{1,\perp}) = C(X)$$

Q6) Prove Theorem 6

$$\pi_X = \pi_{\mathbf{1}} + \pi_{X_{1,\perp}}$$

$$X = [\mathbf{1} \ X_1] \quad , \quad X^T X = \begin{bmatrix} \mathbf{1}^T \mathbf{1} & \mathbf{1}^T X_1 \\ X_1^T \mathbf{1} & X_1^T X_1 \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad , \quad A^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ -A_{22}^{-1} A_{21} A_{11}^{-1} & A_{22}^{-1} A_{21} A_{11}^{-1} A_{12} A_{22}^{-1} + A_{22}^{-1} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11}^{-1} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1} + A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ -A_{22}^{-1} A_{21} A_{11}^{-1} & A_{22}^{-1} \end{bmatrix}$$

$$= \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$A_{22}^{-1} = A_{22}^{-1} A_{11}^{-1} A_{12}$$

$$A_{22}^{-1} = X_1^T X_1 - X_1^T \mathbf{1} (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T X_1 = X_1^T (I - \pi_{\mathbf{1}}) X_1$$

(... too complicated)

Let A be an arbitrary vector/matrix

$$(\pi_{\mathbf{1}} + \pi_{X_{1,\perp}})A = \pi_{\mathbf{1}} A + \pi_{X_{1,\perp}} A$$

: projection of A onto $C(\mathbf{1})$ + projection of A onto $C(X_{1,\perp})$

$\because C(\mathbf{1})$ and $C(X_{1,\perp})$ are orthogonal,

$\Rightarrow$ projection of A onto $(C(\mathbf{1}) + C(X_{1,\perp})) = C(X)$

$\Rightarrow \pi_X A$

$$\therefore \pi_X = \pi_{\mathbf{1}} + \pi_{X_{1,\perp}}$$

Q7) Prove Theorem 7.

Given that $X = [\mathbf{1} \ X_1]$, $X_1 = [x_1 \ \dots \ x_p]$

$$\beta = (\beta_0 \ \gamma_1^T)^T, \ (\beta_0, \dots, \beta_p)^T$$

Show the below:

$$\hat{\beta}_0 = \bar{y} + \frac{1}{n} \mathbf{1}^T X_1 \hat{\gamma}_1$$

$$\hat{\gamma}_1 = (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y$$

$$X \hat{\beta} = \pi_X y = (\pi_{\mathbf{1}} + \pi_{X_{1,\perp}}) y = \pi_{\mathbf{1}} y + \pi_{X_{1,\perp}} y$$

$$X \hat{\beta} = [\mathbf{1} \ X_1] \begin{pmatrix} \hat{\beta}_0 \\ \hat{\gamma}_1 \end{pmatrix} = \mathbf{1} \hat{\beta}_0 + X_1 \hat{\gamma}_1 = \mathbf{1} \hat{\beta}_0 + \pi_{\mathbf{1}} X_1 \hat{\gamma}_1 + X_{1,\perp} \hat{\gamma}_1$$

$$\therefore \mathbf{1} \hat{\beta}_0 + \pi_{\mathbf{1}} X_1 \hat{\gamma}_1 = \pi_{\mathbf{1}} y \quad \text{and} \quad X_{1,\perp} \hat{\gamma}_1 = \pi_{X_{1,\perp}} y$$

$$\therefore \hat{\gamma}_1 = (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y \quad \text{and} \quad \hat{\beta}_0 = (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T y - (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T X_1 \hat{\gamma}_1$$

$$= \frac{1}{n} \mathbf{1}^T y - \frac{1}{n} \mathbf{1}^T X_1 \hat{\gamma}_1$$

$$= \bar{y} - \frac{1}{n} \mathbf{1}^T X_1 \hat{\gamma}_1$$

Q8) Prove Theorem 8

$$E[\hat{\beta}_0] = \beta_0 \quad \text{and} \quad E[\hat{\gamma}_1] = \gamma_1$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \left(\frac{\mathbf{1}^T X_1}{n} \right) (X_{1,\perp}^T X_{1,\perp})^{-1} \left(\frac{X_1^T \mathbf{1}}{n} \right) \right]$$

$$\text{Var}(\hat{\gamma}_1) = \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1}$$

$$E[\hat{\gamma}_1] = E[(X_{1,\perp}^T X_1)^{-1} X_{1,\perp}^T y] = (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T (\mathbf{1} \beta_0 + X_1 \gamma_1)$$

$$= (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T (\mathbf{1} \beta_0 + \pi_{\mathbf{1}} X_1 \gamma_1 + X_{1,\perp} \gamma_1)$$

$$= (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T X_{1,\perp} \gamma_1$$

$$= \gamma_1$$

$$E[\hat{\beta}_0] = E\left[\frac{1}{n} \mathbf{1}^T y - \frac{1}{n} \mathbf{1}^T X_1 \hat{\gamma}_1 \right] = \frac{1}{n} \mathbf{1}^T [\mathbf{1} \beta_0 + X_1 \gamma_1] - \frac{1}{n} \mathbf{1}^T X_1 \gamma_1$$

$$= \frac{1}{n} \cdot n \beta_0 + \frac{1}{n} \mathbf{1}^T X_1 \gamma_1 - \frac{1}{n} \mathbf{1}^T X_1 \gamma_1$$

$$= \beta_0$$

$$\text{Var}(\hat{\gamma}_1) = \text{Var}[(X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y]$$

$$= (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T \text{Var}(y) X_{1,\perp} (X_{1,\perp}^T X_{1,\perp})^{-1}$$

$$= (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T \sigma^2 I_n X_{1,\perp} (X_{1,\perp}^T X_{1,\perp})^{-1}$$

$$= \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1}$$

$$\text{Var}(\hat{\beta}_0) = \text{Var}\left(\frac{1}{n} \mathbf{1}^T y - \frac{1}{n} \mathbf{1}^T X_1 \hat{\gamma}_1 \right)$$

$$= \text{Var}\left(\frac{1}{n} \mathbf{1}^T y \right) + \text{Var}\left(\frac{1}{n} \mathbf{1}^T X_1 \hat{\gamma}_1 \right) - \text{Cov}\left(\frac{1}{n} \mathbf{1}^T y, \frac{1}{n} \mathbf{1}^T X_1 \hat{\gamma}_1 \right)$$

$$\neq \text{Cov}\left(\frac{1}{n} \mathbf{1}^T y, \frac{1}{n} \mathbf{1}^T X_1 (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y \right) = \frac{1}{n^2} \mathbf{1}^T \text{Cov}(y, y) X_{1,\perp} (X_{1,\perp}^T X_{1,\perp})^{-1} X_1^T \mathbf{1}$$

$$= 0 \quad (\because \mathbf{1}^T X_{1,\perp} = 0)$$

$$\text{Var}\left(\frac{1}{n} \mathbf{1}^T y \right) = \frac{1}{n^2} \mathbf{1}^T \text{Var}(y) \mathbf{1} = \sigma^2 \frac{1}{n^2} \mathbf{1}^T \mathbf{1} = \frac{\sigma^2}{n}$$

$$\therefore \text{Var}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \left(\frac{\mathbf{1}^T X_1}{n} \right) (X_{1,\perp}^T X_{1,\perp})^{-1} \left(\frac{X_1^T \mathbf{1}}{n} \right) \right]$$

Q9) What is the matrix representation of the following estimates :

- (a) $E[\hat{\beta}_0]$
 (b) $E[\hat{\gamma}_1]$
 (c) $\text{Var}(\hat{\beta}_0)$
 (d) $\text{Var}(\hat{\gamma}_1)$

(a) $E[\hat{\beta}_0] = \beta = (1 \ 0^T) \beta$

(b) $E[\hat{\gamma}_1] = (0 \ I_p) \beta$

(c) $\text{Var}(\hat{\beta}_0) = \sigma^2 (1 \ 0^T) (X^T X)^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

(d) $\text{Var}(\hat{\gamma}_1) = \sigma^2 (0 \ I_p) (X^T X)^{-1} \begin{pmatrix} 0 \\ I_p \end{pmatrix}$

$$\text{Var}(\hat{\beta}_0) = (1 \ 0^T) \text{Var}(\hat{\beta}) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \sigma^2 (1 \ 0^T) (X^T X)^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{Var}(\hat{\gamma}_1) = \sigma^2 (0 \ I_p) (X^T X)^{-1} \begin{pmatrix} 0 \\ I_p \end{pmatrix}$$

Q10) Prove Corollary 10

(a) $E[\hat{\mu}_1] = \mu_1$

(b) $E[\hat{y}_2] = E[y_2]$

(c) $E[\hat{\epsilon}_1] = E[\epsilon_1]$

(d) $E[\hat{\sigma}^2] = \sigma^2$

(e) $\text{Var}(\hat{\mu}_1) = \sigma^2 \left[\frac{1}{n} + (z_1^T - \frac{1^T X_1}{n}) (X_{1+}^T X_{1+})^{-1} (z_1 - \frac{X_1^T 1}{n}) \right]$

(f) $\text{Var}(\hat{y}_2) = \sigma^2 \left[\frac{1}{n} + (z^T - \frac{1^T X}{n}) (X_{1+}^T X_{1+})^{-1} (z - \frac{X_1^T 1}{n}) \right]$

(g) $\text{Var}(\hat{\epsilon}_1) = \sigma^2 \left[1 - \frac{1}{n} - (z_1^T - \frac{1^T X_1}{n}) (X_{1+}^T X_{1+})^{-1} (z_1 - \frac{X_1^T 1}{n}) \right]$

(a) $\hat{\mu}_1 = \hat{\beta}_0 + z_1^T \hat{\gamma}_1, \therefore E[\hat{\mu}_1] = E[\hat{\beta}_0 + z_1^T \hat{\gamma}_1] = \beta_0 + z_1^T \gamma_1 = \mu_1$

(b) $\hat{y}_2 = \hat{\beta}_0 + z^T \hat{\gamma}_1, \therefore E[\hat{y}_2] = E[\hat{\beta}_0 + z^T \hat{\gamma}_1] = \beta_0 + z^T \gamma_1 = E[y_2]$

(c) $\hat{\epsilon}_1 = y_1 - \hat{y}_1, E[\hat{\epsilon}_1] = E[y_1] - E[\hat{y}_1] = \mu_1 - E[\hat{\mu}_1] = 0 = E[\epsilon_1]$

(d) $\hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i=1}^n \hat{\epsilon}_i^2 = \frac{1}{n-p-1} \|\hat{\epsilon}\|^2 = \frac{1}{n-p-1} \| (I_n - \Pi_X) y \|^2 = \frac{1}{n-p-1} y^T (I_n - \Pi_X) y$
 $y^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) y \sim \chi^2(r; \delta), r = \text{rank} \left(\frac{I_n - \Pi_X}{\sigma^2} \right), \delta = \mu^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) \mu$

$\mu = E[y] = X\beta, \mu^T (I_n - \Pi_X) \mu = \beta^T X^T (I_n - \Pi_X) X \beta = \beta^T X^T (X\beta - X(X^T X)^{-1} X^T X \beta) = 0$
 $\therefore \delta = 0$

$r = \text{rank} \left(\frac{I_n - \Pi_X}{\sigma^2} \right) = \text{rank}(I_n - \Pi_X) = \text{tr}(I_n - \Pi_X) = \text{tr}(I_n) - \text{tr}(X(X^T X)^{-1} X^T)$
 $= \text{tr}(I_n) - \text{tr}(X^T X)^{-1} X^T X = n - (p+1) = n - p - 1$

$A \Sigma = \left(\frac{I_n - \Pi_X}{\sigma^2} \right) \cdot \sigma^2 I_n = I_n - \Pi_X : \text{idempotent}$

$\therefore y^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) y \sim \chi^2(n-p-1)$

$\Rightarrow E[y^T (I_n - \Pi_X) y] = \sigma^2 (n-p-1)$

$\Rightarrow E[\hat{\sigma}^2] = E \left[\frac{1}{n-p-1} y^T (I_n - \Pi_X) y \right] = \sigma^2$

(e) Approach #1

$$\begin{aligned} \text{Var}(\hat{\mu}_1) &= \text{Var}(\hat{y}_{z_1}) = \text{Var}(\hat{\beta}_0 + z_1^T \hat{\gamma}_1) \\ &= \text{Var}(\hat{\beta}_0) + \text{Var}(z_1^T \hat{\gamma}_1) + 2 \text{Cov}(\hat{\beta}_0, z_1^T \hat{\gamma}_1) \\ &= \underbrace{\text{Var} \left(\frac{1}{n} 1^T y - \frac{1}{n} 1^T X_1 \hat{\gamma}_1 \right)}_{\textcircled{1}} + \underbrace{z_1^T \text{Var}(\hat{\gamma}_1) z_1}_{\textcircled{2}} + \underbrace{2 \text{Cov}(\hat{\beta}_0, z_1^T \hat{\gamma}_1)}_{\textcircled{3}} \end{aligned}$$

① $\text{Var} \left(\frac{1}{n} 1^T y - \frac{1}{n} 1^T X_1 \hat{\gamma}_1 \right)$

$$= \frac{1}{n^2} 1^T \text{Var}(y) 1 + \left(\frac{1^T X_1}{n} \right) \text{Var}(\hat{\gamma}_1) \left(\frac{X_1^T 1}{n} \right) - 2 \text{Cov} \left(\frac{1}{n} 1^T y, \frac{1}{n} 1^T X_1 \hat{\gamma}_1 \right)$$

$$= \frac{1}{n^2} 1^T \text{Var}(y) 1 + \left(\frac{1^T X_1}{n} \right) \text{Var}(\hat{\gamma}_1) \left(\frac{X_1^T 1}{n} \right) \because \text{Cov} \left(\frac{1}{n} 1^T y, \frac{1}{n} 1^T X_1 \hat{\gamma}_1 \right) = 0$$

$1^T y \in C(1), \hat{\gamma}_1 \in C(X_{1+})$

② $z_1^T \text{Var}(\hat{\gamma}_1) z_1 = z_1^T \left(\sigma^2 (X_{1+}^T X_{1+})^{-1} \right) z_1 = \sigma^2 z_1^T (X_{1+}^T X_{1+})^{-1} z_1$

③ $2 \text{Cov}(\hat{\beta}_0, z_1^T \hat{\gamma}_1) = 2 \text{Cov}(\hat{\beta}_0, \hat{\gamma}_1) z_1 = 0 \because \hat{\beta}_0 \in C(1), \hat{\gamma}_1 \in C(X_{1+})$

$$\therefore \text{Var}(\hat{\mu}_1) = \sigma^2 \left(\frac{1}{n} + \left(\frac{1^T X_1}{n} \right) (X_{1+}^T X_{1+})^{-1} \left(\frac{X_1^T 1}{n} \right) + z_1^T (X_{1+}^T X_{1+})^{-1} z_1 \right)$$

Approach #2

$$\begin{aligned} \text{Var}(\hat{\mu}_1) &= \text{Var}(\hat{\beta}_0 + z_1^T \hat{\gamma}_1) = \text{Var} \left(\frac{1}{n} 1^T y - \frac{1}{n} 1^T X_1 \hat{\gamma}_1 + z_1^T \hat{\gamma}_1 \right) \\ &= \text{Var} \left(\frac{1}{n} 1^T y \right) + \text{Var} \left(\left(z_1^T - \frac{1}{n} 1^T X_1 \right) \hat{\gamma}_1 \right) \\ &= \frac{1}{n^2} 1^T \text{Var}(y) 1 + (z_1^T - \frac{1}{n} 1^T X_1) \text{Var}(\hat{\gamma}_1) (z_1 - \frac{1}{n} X_1^T 1) \\ &= \sigma^2 \left(\frac{1}{n} + (z_1^T - \frac{1}{n} 1^T X_1) (X_{1+}^T X_{1+})^{-1} (z_1 - \frac{1}{n} X_1^T 1) \right) \end{aligned}$$

(f) $\text{Var}(\hat{y}_2) = \text{Var}(\hat{\beta}_0 + z^T \hat{\gamma}_1)$
 $= \text{Var} \left(\frac{1}{n} 1^T y - \frac{1}{n} 1^T X_1 \hat{\gamma}_1 - z^T \hat{\gamma}_1 \right)$
 $= \sigma^2 \left(\frac{1}{n} + (z^T - \frac{1}{n} 1^T X_1) (X_{1+}^T X_{1+})^{-1} (z - \frac{1}{n} X_1^T 1) \right)$

(g) $\text{Var}(\hat{\epsilon}_1) = \text{Var}(y_1 - \hat{y}_1) = \text{Var}(y_1) + \text{Var}(\hat{y}_1) - 2 \text{Cov}(y_1, \hat{y}_1)$
 $= \text{Var}(y_1) + \text{Var}(\hat{y}_1) - 2 \text{Cov}(\hat{\epsilon}_1 + \hat{y}_1, \hat{y}_1)$
 $= \text{Var}(y_1) + \text{Var}(\hat{y}_1) - 2 \text{Var}(\hat{y}_1) - 2 \text{Cov}(\hat{\epsilon}_1, \hat{y}_1)$
 $= \text{Var}(y_1) - \text{Var}(\hat{y}_1) (\because \text{Cov}(\hat{\epsilon}_1, \hat{y}_1) = 0)$
 $= \sigma^2 \left(1 - \frac{1}{n} - (z_1^T - \frac{1}{n} 1^T X_1) (X_{1+}^T X_{1+})^{-1} (z_1 - \frac{1}{n} X_1^T 1) \right)$

$$\begin{aligned} + \text{Cov}(\hat{\epsilon}_1, \hat{y}_1) &= \text{Cov}((I_n - \Pi_X) y, \Pi_X y) \\ &= (I_n - \Pi_X) \text{Var}(y) \Pi_X \\ &= \sigma^2 (I_n - \Pi_X) \Pi_X \\ &= 0 \end{aligned}$$

Q11) Derive the following:

$$(a) \text{Var}(\hat{\mu}_1) = \sigma^2 (1 \text{ } z_1^T) (X^T X)^{-1} \begin{pmatrix} 1 \\ z_1 \end{pmatrix}$$

$$(b) \text{Var}(\hat{y}_2) = \sigma^2 (1 \text{ } z^T) (X^T X)^{-1} \begin{pmatrix} 1 \\ z \end{pmatrix}$$

[Solution]:

$$(a) \text{Var}(\hat{\mu}_1) = \text{Var}(\hat{\beta}_0 + z_1^T \hat{\gamma}_1) = \text{Var} \left[\begin{pmatrix} 1 & z_1^T \end{pmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\gamma}_1 \end{bmatrix} \right]$$

$$= (1 \text{ } z_1^T) \text{Var}(\hat{\beta}_0) \begin{pmatrix} 1 \\ z_1 \end{pmatrix}$$

$$= (1 \text{ } z_1^T) \text{Var} \left(\begin{pmatrix} X_1^T X_1 & X_1^T y \\ X_1^T y & X_1^T X_1 \end{pmatrix} \begin{pmatrix} 1 \\ z_1 \end{pmatrix} \right)$$

$$= (1 \text{ } z_1^T) (X_1^T X_1)^{-1} X_1^T \text{Var}(y) X_1 (X_1^T X_1)^{-1} \begin{pmatrix} 1 \\ z_1 \end{pmatrix}$$

$$= \sigma^2 (1 \text{ } z_1^T) (X_1^T X_1)^{-1} \begin{pmatrix} 1 \\ z_1 \end{pmatrix}$$

$$(b) \text{Var}(\hat{y}_2) : z_1 \Rightarrow z \text{ in } \text{Var}(\hat{\mu}_1)$$

$$= \sigma^2 (1 \text{ } z^T) (X^T X)^{-1} \begin{pmatrix} 1 \\ z \end{pmatrix}$$

ANOVA (Analysis of Variance)

Q12) Prove Theorem 11

$$SST = \|y - \bar{y}\|^2 = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SSE = \|y - \hat{y}\|^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSR = \|\hat{y} - \bar{y}\|^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

[Solution]

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2$$

$$= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) + \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) = [(I_n - \Pi_X) y]^T [(\Pi_X - \Pi_{\bar{y}}) y]$$

$$= y^T (I_n - \Pi_X)(\Pi_X - \Pi_{\bar{y}}) y$$

$$I_n - \Pi_X \in C(X)^\perp \text{ and } \Pi_X - \Pi_{\bar{y}} \in C(X)$$

$$\therefore (I_n - \Pi_X)(\Pi_X - \Pi_{\bar{y}}) = 0$$

$$\therefore SST = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = SSE + SSR$$

Q13) Prove Theorem 12

$$(a) SST = SSR + SSE$$

$$SST = y^T (I_n - \Pi_{\bar{y}}) y$$

$$SSE = y^T (I_n - \Pi_X) y$$

$$SSR = y^T (\Pi_X - \Pi_{\bar{y}}) y$$

(b) For the regression sum of square,

$$SSR = y^T \Pi_{X_{\perp}} y = \hat{\gamma}_1^T X_{1,\perp}^T X_{1,\perp} \hat{\gamma}_1$$

(a)

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = [(I_n - \Pi_{\bar{y}}) y]^T [(I_n - \Pi_{\bar{y}}) y] \\ = y^T (I_n - \Pi_{\bar{y}}) y$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = [(I_n - \Pi_X) y]^T [(I_n - \Pi_X) y] \\ = y^T (I_n - \Pi_X) y$$

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = [(\Pi_X - \Pi_{\bar{y}}) y]^T [(\Pi_X - \Pi_{\bar{y}}) y] \\ = y^T (\Pi_X - \Pi_{\bar{y}}) y$$

$$(b) SSR = y^T (\Pi_X - \Pi_{\bar{y}}) y = y^T (\Pi_{\bar{y}} + \Pi_{X_{\perp}} - \Pi_{\bar{y}}) y$$

$$= y^T \Pi_{X_{\perp}} y = (\Pi_{X_{\perp}} y)^T (\Pi_{X_{\perp}} y) \\ = (X_{1,\perp} (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y)^T (X_{1,\perp} (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y) \\ = (X_{1,\perp}^T \hat{\gamma}_1)^T (X_{1,\perp}^T \hat{\gamma}_1) \\ = \hat{\gamma}_1^T X_{1,\perp}^T X_{1,\perp} \hat{\gamma}_1$$

Q14) Define the coefficient of Determination R^2 in terms of SSR and SST

$$R^2 = \frac{SSR}{SST}$$

Q15) Prove Theorem 14

$$(a) R^2 = 0 \Leftrightarrow \hat{\beta}_1 = \dots = \hat{\beta}_p = 0$$

$$(b) R^2 = 0 \Leftrightarrow y = X\hat{\beta}$$

(c) R^2 = Square of the sample correlation coefficient between y and $\hat{y}$.

$$R^2 = \frac{\left[\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{y}) \right]^2}{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{y})^2}, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$(a) R^2 = 0 \Leftrightarrow SSR = 0 \Leftrightarrow \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = 0 \Leftrightarrow \hat{y}_i = \bar{y} \quad (i=1, 2, \dots, n) \\ y^T (\Pi_X - \Pi_{\bar{y}}) y = 0$$

$$\|(\Pi_X - \Pi_{\bar{y}}) y\|^2 = 0 \Leftrightarrow \Pi_X y = \Pi_{\bar{y}} y$$

$$\Pi_X = \Pi_{\bar{y}} + \Pi_{X_{\perp}}, \quad \therefore \Pi_{\bar{y}} y + \Pi_{X_{\perp}} y = \Pi_{\bar{y}} y \Rightarrow \Pi_{X_{\perp}} y = 0 \\ \Rightarrow X_{1,\perp} (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y = 0 \Leftrightarrow (X_{1,\perp}^T X_{1,\perp})^{-1} (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y = 0 \\ \Leftrightarrow (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y = 0 \\ \Leftrightarrow \hat{\gamma}_1 = 0 \\ \therefore \hat{\beta}_1 = \dots = \hat{\beta}_p = 0$$

$$(b) R^2 = 1 \Leftrightarrow SSR = SST \Leftrightarrow \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\Leftrightarrow y^T (\Pi_X - \Pi_{\perp}) y = y^T (I_n - \Pi_{\perp}) y$$

$$\Leftrightarrow \|(\Pi_X - \Pi_{\perp}) y\|^2 = \|(I_n - \Pi_{\perp}) y\|^2$$

$$\therefore \Pi_X y = y \Leftrightarrow X\hat{\beta} = y$$

$$(c) \frac{[\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{y})]^2}{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{y})^2}$$

$$\text{numerator: } [(I_n - \Pi_{\perp}) y]^T [(\Pi_X - \Pi_{\perp}) y]$$

$$= y^T (I_n - \Pi_{\perp}) (\Pi_X - \Pi_{\perp}) y$$

$$= y^T (\Pi_X - \Pi_{\perp} - \Pi_{\perp} + \Pi_{\perp}) y$$

$$= y^T \Pi_X y = SSR$$

$$\text{denominator: } \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = [(\Pi_X - \Pi_{\perp}) y]^T [(\Pi_X - \Pi_{\perp}) y]$$

$$= y^T (\Pi_X - \Pi_{\perp}) \Pi_X y$$

$$= y^T (\Pi_X - \Pi_{\perp}) y = SSR$$

$$\therefore \frac{[\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{y})]^2}{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{y})^2} = \frac{SSR^2}{SST \cdot SSR} = \frac{SSR}{SST} = R^2$$

Distribution of Estimators:

Q16) Prove Theorem 16

$$(a) \hat{\gamma}_1 \sim N_p(\gamma_1, \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1})$$

$$(b) (n-p-1) \hat{\sigma}^2 / \sigma^2 \sim \chi^2(n-p-1)$$

$$(c) \hat{\gamma}_1 \text{ and } \hat{\sigma}^2 \text{ are independent}$$

$$(a) E[\hat{\gamma}_1] = \gamma_1 \text{ and } \text{Var}(\hat{\gamma}_1) = \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1}$$

$$\therefore \hat{\gamma}_1 \sim N(\gamma_1, \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1})$$

$$(b) (n-p-1) \hat{\sigma}^2 / \sigma^2 = \frac{(n-p-1)}{\sigma^2} \cdot \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \frac{1}{\sigma^2} y^T (I_n - \Pi_X) y = y^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) y \sim \chi^2(r; \delta), \quad \delta = \mu^T A \mu, \quad \mu = E[y]$$

$$\Leftrightarrow A \Sigma \text{ is idempotent}$$

$$A = \left(\frac{I_n - \Pi_X}{\sigma^2} \right)$$

$$A \Sigma = \left(\frac{I_n - \Pi_X}{\sigma^2} \right) \sigma^2 I_n = I_n - \Pi_X : \text{idempotent.}$$

$$\therefore y^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) y \sim \chi^2(n-p-1) \quad \therefore \text{rank} \left(\frac{I_n - \Pi_X}{\sigma^2} \right) = \text{trace}(I_n - \Pi_X)$$

$$= \text{tr}(I_n) - \text{tr}(X(X^T X)^{-1} X^T)$$

$$= \text{tr}(I_n) - \text{tr}(X^T X (X^T X)^{-1})$$

$$= n - (p+1) = n-p-1$$

$$\delta = \mu^T A \mu = (X\beta)^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) (X\beta)$$

$$(I_n - \Pi_X) X = X - X(X^T X)^{-1} X^T X = 0 \quad \therefore \delta = 0$$

$$(c) \hat{\gamma}_1 = (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T y \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n-p-1} y^T (I_n - \Pi_X) y$$

$$\hat{\gamma}_1 \in C(X_{1,\perp}) \subset C(X) \text{ and } \hat{\sigma}^2 \in C(X)^\perp$$

$$\therefore \hat{\gamma}_1 \text{ and } \hat{\sigma}^2 \text{ are orthogonal}$$

$$A \Sigma B = [(X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T] \left[\sigma^2 I_n \right] \left[\frac{I_n - \Pi_X}{\sigma^2} \right] = 0$$

$$\therefore A y = \hat{\gamma}_1 \text{ and } y^T B y = (n-p-1) \hat{\sigma}^2 \text{ are independent}$$

$$\Rightarrow \hat{\gamma}_1 \text{ and } \hat{\sigma}^2 \text{ are independent.}$$

Q17) Prove Remark in Chapter 53

$$(a) SSE / \sigma^2 \sim \chi^2(n-p-1)$$

$$(b) SSR / \sigma^2 \sim \chi^2(p; \delta)$$

$$(a) SSE / \sigma^2 = y^T (I_n - \Pi_X) y / \sigma^2 = y^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) y = y^T B y$$

$$B \Sigma = \left(\frac{I_n - \Pi_X}{\sigma^2} \right) \cdot \sigma^2 I_n = I_n - \Pi_X : \text{idempotent}$$

$$\therefore y^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) y \sim \chi^2(r; \delta)$$

$$r = \text{rank} \left(\frac{I_n - \Pi_X}{\sigma^2} \right) = \text{rank}(I_n - \Pi_X) = n-p-1$$

$$\delta = \mu^T A \mu = E[y]^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) E[y] = \beta^T X^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) X \beta = 0 \quad \therefore \langle I_n - \Pi_X, X \rangle = 0$$

$$(b) SSR / \sigma^2 = y^T \left(\frac{\Pi_X - \Pi_{\perp}}{\sigma^2} \right) y = y^T A y$$

$$A \Sigma = \left(\frac{\Pi_X - \Pi_{\perp}}{\sigma^2} \right) \cdot \sigma^2 I_n = \Pi_X - \Pi_{\perp} : \text{idempotent}$$

$$\therefore y^T \left(\frac{\Pi_X - \Pi_{\perp}}{\sigma^2} \right) y \sim \chi^2(r; \delta)$$

$$r = \text{rank}(\Pi_X - \Pi_{\perp}) = \text{trace}(\Pi_X - \Pi_{\perp}) = \text{trace}(\Pi_X) - \text{trace}(\Pi_{\perp})$$

$$= (p+1) - 1 = p$$

$$\delta = \mu^T A \mu = E[y]^T \left(\frac{\Pi_X - \Pi_{\perp}}{\sigma^2} \right) E[y] = (X\beta)^T \left(\frac{\Pi_X - \Pi_{\perp}}{\sigma^2} \right) (X\beta)$$

$$= (X\beta)^T \left(\frac{\Pi_{X_{1,\perp}} + \Pi_{\perp} - \Pi_{\perp}}{\sigma^2} \right) (X\beta)$$

$$= \sigma^{-2} (X_{1,\perp}^T X_{1,\perp})^{-1} \Pi_{X_{1,\perp}} (X_{1,\perp}^T X_{1,\perp})$$

$$= \sigma^{-2} X_{1,\perp}^T (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}$$

$$\Rightarrow \hat{\gamma}_1 \text{ and } \hat{\sigma}^2 \text{ are independent.}$$

Q18) Prove Theorem 17

$$H_0: \gamma_1 = \gamma_1^* \text{ vs } H_1: \gamma_1 \neq \gamma_1^*$$

$$F = \frac{(\hat{\gamma}_1 - \gamma_1^*)^T X_{1,\perp}^T X_{1,\perp} (\hat{\gamma}_1 - \gamma_1^*) / p}{SSE / (n-p-1)}$$

Under $H_0: F \sim F(p, n-p-1)$

$$\text{Under } H_1: F \sim F(p, n-p-1; \delta), \delta = \frac{(\gamma_1 - \gamma_1^*)^T X_{1,\perp}^T X_{1,\perp} (\gamma_1 - \gamma_1^*)}{\sigma^2}$$

$$\begin{aligned} \hat{\gamma}_1 &\sim N(\gamma_1, \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1}) \\ (\hat{\gamma}_1 - \gamma_1) / \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1/2} &\sim N(0, I_p) \\ \therefore (\hat{\gamma}_1 - \gamma_1^*)^T X_{1,\perp}^T X_{1,\perp} (\hat{\gamma}_1 - \gamma_1^*) / \sigma^2 &\sim \chi^2(p) \end{aligned}$$

$$SSE = y^T (I - \Pi_X) y, \quad SSE / \sigma^2 = y^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) y \sim \chi^2(n-p-1)$$

$$\therefore \frac{(\hat{\gamma}_1 - \gamma_1^*)^T X_{1,\perp}^T X_{1,\perp} (\hat{\gamma}_1 - \gamma_1^*) / p}{y^T (I_n - \Pi_X) y / (n-p-1)} \sim \frac{V_1 / n_1}{V_2 / n_2}, \quad V_1 \sim \chi^2(n_1), V_2 \sim \chi^2(n_2)$$

$\therefore$ under $H_0, F \sim F(p, n-p-1)$

$$\text{under } H_1, \hat{\gamma}_1 - \gamma_1^* \sim N(\gamma_1 - \gamma_1^*, \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1})$$

$$\therefore (\hat{\gamma}_1 - \gamma_1^*)^T X_{1,\perp}^T X_{1,\perp} (\hat{\gamma}_1 - \gamma_1^*) / \sigma^2 \sim \chi^2(p; \delta), \delta = (\gamma_1 - \gamma_1^*)^T X_{1,\perp}^T X_{1,\perp} (\gamma_1 - \gamma_1^*) / \sigma^2$$

$$\therefore F \sim F(p, n-p-1; \delta), \delta = \frac{(\gamma_1 - \gamma_1^*)^T X_{1,\perp}^T X_{1,\perp} (\gamma_1 - \gamma_1^*)}{\sigma^2} \quad \square$$

Q19) Prove Theorem 18

$$H_0: \gamma_1 = 0 \text{ vs } H_1: \gamma_1 \neq 0$$

$$F = \frac{\hat{\gamma}_1^T X_{1,\perp}^T X_{1,\perp} \hat{\gamma}_1 / p}{SSE / (n-p-1)} \quad \begin{aligned} \text{Under } H_0: F &\sim F(p, n-p-1) \\ \text{Under } H_1: F &\sim F(p, n-p-1; \delta), \delta = \frac{\gamma_1^T X_{1,\perp}^T X_{1,\perp} \gamma_1}{\sigma^2} \end{aligned}$$

insert γ_1^* into Theorem 16 (previous question)

Q20) Prove Theorem 19

$$H_0: \beta_j = \beta_j^* \text{ vs } H_1: \beta_j \neq \beta_j^*$$

for some $\beta_j^* \in \mathbb{R}$. Consider a test statistic T defined as below

$$T = \frac{\hat{\beta}_j - \beta_j^*}{\sqrt{\hat{\sigma}^2 C_{jj}}}$$

Let $e_j = [0 \dots 1 \dots 0]^T$ 1 is in the j^{th} row

$$e_j^T \hat{\gamma}_1 \sim N(e_j^T \gamma_1, e_j^T \sigma^2 (X_{1,\perp}^T X_{1,\perp})^{-1} e_j)$$

$$\Rightarrow \hat{\beta}_j \sim N(\beta_j, \sigma^2 C_{jj}), \quad C_{jj}: j^{\text{th}} \text{ diagonal entry of } (X_{1,\perp}^T X_{1,\perp})^{-1}$$

$$(\hat{\beta}_j - \beta_j^*) / \sqrt{\sigma^2 C_{jj}} \sim N(0, I_1)$$

$$\frac{\hat{\beta}_j - \beta_j^*}{\sqrt{\hat{\sigma}^2 C_{jj}}} = \frac{\hat{\beta}_j - \hat{\beta}_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \Bigg/ \sqrt{\frac{(n-p-1)\hat{\sigma}^2}{\sigma^2}} \Bigg/ (n-p-1) = \frac{Z}{\sqrt{V/(n-p-1)}}, \quad \begin{aligned} Z &\sim N(0, 1) \\ V &\sim \chi^2(n-p-1) \end{aligned}$$

$$\therefore \frac{\hat{\beta}_j - \beta_j^*}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t(n-p-1)$$

Q21) Prove Theorem 20

$$H_0: \beta_0 = \beta_0^* \text{ vs } H_1: \beta_0 \neq \beta_0^*$$

for some constant $\beta_0^* \in \mathbb{R}$. Consider the test statistic T below:

$$T = \frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2 D_{11}}}$$

under $H_0, T \sim t(n-p-1)$

D_{11} : 1st diagonal entry of $(X^T X)^{-1}$

$$D_{11} = \left[\frac{1}{n} + \frac{(X_{1,\perp}^T X_{1,\perp})^{-1}}{n} \right]^{-1} \left[\frac{X_{1,\perp}^T X_{1,\perp}}{n} \right] \quad \downarrow \text{at } 2H \text{ II-10-2}$$

$$\begin{aligned} \hat{\beta}_0 &\sim N(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{(X_{1,\perp}^T X_{1,\perp})^{-1}}{n} \right) \left(\frac{X_{1,\perp}^T X_{1,\perp}}{n} \right)) \\ &\stackrel{d}{=} N(\beta_0, \sigma^2 D_{11}) \end{aligned}$$

$$\therefore \text{under } H_0, \frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2 D_{11}}} \sim N(0, 1)$$

$$\frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2 D_{11}}} = \frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\sigma^2 D_{11}}} \Bigg/ \sqrt{\frac{(n-p-1)\hat{\sigma}^2}{\sigma^2}} \Bigg/ (n-p-1) = \frac{Z}{\sqrt{V/(n-p-1)}}, \quad \begin{aligned} Z &\sim N(0, 1) \\ V &\sim \chi^2(n-p-1) \end{aligned}$$

$$\therefore (\hat{\beta}_0 - \beta_0^*) / \sqrt{\hat{\sigma}^2 D_{11}} \sim t(n-p-1)_0$$

CONFIDENCE INTERVAL

Q22) Prove Theorem 22

The $(1-\alpha)100\%$ CI for β_j and β_0 are:

$$\beta_j: \hat{\beta}_j \pm t_{\alpha/2, (n-p-1)} \sqrt{\hat{\sigma}^2 C_{jj}}$$

$$\beta_0: \hat{\beta}_0 \pm t_{\alpha/2, (n-p-1)} \sqrt{\hat{\sigma}^2 D_{11}}$$

$$\begin{aligned} \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} &\sim t(n-p-1) \quad \therefore P(|(\hat{\beta}_j - \beta_j) / \sqrt{\hat{\sigma}^2 C_{jj}}| > t_{\alpha/2}) = \alpha \\ &\Rightarrow P(\hat{\beta}_j - \beta_j / \sqrt{\hat{\sigma}^2 C_{jj}} > t_{\alpha/2}, \hat{\beta}_j - \beta_j / \sqrt{\hat{\sigma}^2 C_{jj}} < -t_{\alpha/2}) \end{aligned}$$

$$\begin{aligned} \therefore \text{CI: } -t_{\alpha/2} &< (\hat{\beta}_j - \beta_j) / \sqrt{\hat{\sigma}^2 C_{jj}} < t_{\alpha/2} \\ &= \hat{\beta}_j - t_{\alpha/2} \sqrt{\hat{\sigma}^2 C_{jj}} < \beta_j < \hat{\beta}_j + t_{\alpha/2} \sqrt{\hat{\sigma}^2 C_{jj}} \end{aligned}$$

$$\therefore \beta_j \text{ CI: } \hat{\beta}_j \pm t_{\alpha/2} \sqrt{\hat{\sigma}^2 C_{jj}}, \quad t_{\alpha/2} = t_{\alpha/2, (n-p-1)}$$

$$\frac{\hat{\beta}_0 - \beta_0}{\sqrt{\hat{\sigma}^2 D_{11}}} \sim t(n-p-1)$$

$$\therefore \text{Similar to above } -t_{\alpha/2, (n-p-1)} < \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\hat{\sigma}^2 D_{11}}} < t_{\alpha/2, (n-p-1)}$$

$$\therefore \text{CI of } \beta_0: \hat{\beta}_0 \pm t_{\alpha/2, (n-p-1)} \sqrt{\hat{\sigma}^2 D_{11}}$$

Q23) Prove Theorem 2.2

$$\text{Let } C_2 = \left[\frac{1}{n} + \left(Z^T - \frac{1^T X_1}{n} \right) (X_{1,\perp}^T X_{1,\perp})^{-1} \left(Z^T - \frac{X_1^T 1}{n} \right) \right]$$

(a) CI of μ_i at $Z_i, i=1, \dots, n$: $\hat{\mu}_i \pm t_{\alpha/2}(n-p-1) \sqrt{\hat{\sigma}^2 C_{ii}}$

(b) CI of y_z at $z \in \mathbb{R}^1$ (y_z : out-of-sample response)

$$\hat{y}_z \pm t_{\alpha/2}(n-p-1) \sqrt{\hat{\sigma}^2 (1 + C_z)}$$

(proof)

(a) $\hat{\mu}_i = \hat{\beta}_0 + Z_i^T \hat{\gamma}_1 \sim N(\mu_i, \text{Var}(\hat{\mu}_i)) \stackrel{d}{=} N(\mu_i, \sigma^2 C_z)$

$$\therefore \frac{\hat{\mu}_i - \mu_i}{\sqrt{\sigma^2 C_z}} \sim N(0, 1)$$

$$\Rightarrow \frac{\hat{\mu}_i - \mu_i}{\sqrt{\hat{\sigma}^2 C_z}} = \frac{\hat{\mu}_i - \mu_i}{\sqrt{\sigma^2 C_z}} \sqrt{\frac{(n-p-1) \hat{\sigma}^2}{\sigma^2}} \sqrt{(n-p-1)} = \frac{Z}{\sqrt{V/(n-p-1)}} \quad Z \sim N(0, 1) \quad V \sim \chi^2(n-p-1)$$

$$\therefore \frac{\hat{\mu}_i - \mu_i}{\sqrt{\hat{\sigma}^2 C_z}} \sim t(n-p-1) \quad \mu_i \text{ CI : } \hat{\mu}_i \pm t_{\alpha/2}(n-p-1) \sqrt{\hat{\sigma}^2 C_z}$$

(b) $y_z \sim N(\beta_0 + Z^T \gamma, \sigma^2)$

$$\hat{y}_z - y_z \sim N(0, \sigma^2 (1 + C_z)) \quad \therefore \hat{y}_z \text{ and } y_z \text{ are independent}$$

$$\frac{\hat{y}_z - y_z}{\sqrt{\sigma^2 (1 + C_z)}} \sim N(0, 1)$$

$$\therefore \frac{\hat{y}_z - y_z}{\sqrt{\hat{\sigma}^2 (1 + C_z)}} = \frac{\hat{y}_z - y_z}{\sqrt{\sigma^2 (1 + C_z)}} \sqrt{\frac{(n-p-1) \hat{\sigma}^2}{\sigma^2}} \sqrt{(n-p-1)} = \frac{Z}{\sqrt{V/(n-p-1)}} \quad Z \sim N(0, 1) \quad V \sim \chi^2(n-p-1)$$

$$\therefore \frac{\hat{y}_z - y_z}{\sqrt{\hat{\sigma}^2 (1 + C_z)}} \sim t(n-p-1) \Rightarrow \text{CI } y_z : \hat{y}_z \pm t_{\alpha/2}(n-p-1) \sqrt{\hat{\sigma}^2 (1 + C_z)}$$

$$* D_{11} = \left[\frac{1}{n} + \left(\frac{1^T X_1}{n} \right)^T (X_{1,\perp}^T X_{1,\perp})^{-1} \left(\frac{X_1^T 1}{n} \right) \right]$$

$$D_{11} = e_1^T (X^T X)^{-1} e_1$$

$$X = \begin{bmatrix} 1 & X_1 \end{bmatrix}, \quad X^T X = \begin{bmatrix} 1^T & 1^T X_1 \\ X_1^T 1 & X_1^T X_1 \end{bmatrix} = \begin{bmatrix} 1^T 1 & 1^T X_1 \\ X_1^T 1 & X_1^T X_1 \end{bmatrix}$$

$$\text{let } P = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} A^{-1} + A^{-1} B (D - C A^{-1} B)^{-1} C A^{-1} & \dots \\ \dots & \dots \end{bmatrix}$$

$$= \begin{bmatrix} (A - B D^{-1} C)^{-1} & \dots \\ \dots & \dots \end{bmatrix}$$

$$\text{we will use } A^{-1} + A^{-1} B (D - C A^{-1} B)^{-1} C A^{-1}$$

$$= \Sigma_{11}^{-1} + \Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1}$$

$$D_{11} = (1^T 1)^{-1} + (1^T 1)^{-1} 1^T X_1 (X_1^T X_1 - X_1^T 1 (1^T 1)^{-1} 1^T X_1)^{-1} X_1^T 1 (1^T 1)^{-1}$$

$$= \frac{1}{n} + \left(\frac{1^T X_1}{n} \right) (X_1^T (I - \Pi_1) X_1)^{-1} \left(\frac{X_1^T 1}{n} \right)$$

$$= \frac{1}{n} + \left(\frac{1^T X_1}{n} \right) (X_{1,\perp}^T X_{1,\perp})^{-1} \left(\frac{X_1^T 1}{n} \right)$$